Intelligent Remote Controlled Antenna Matrix for Shortwave Radio

Professional Bachelor Electronics-ICT
Major Electronics

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Foreword

This bachelor thesis marks the final stage of my studies in Electronics-ICT and represents an important learning experience for me. The project gave me the opportunity to work on a challenging and practical topic that combines RF design, PCB development, measurement techniques, and embedded software.

During this thesis, I learned not only technical skills, but also the importance of making design decisions step by step, based on calculations, testing, and feedback. The project was sometimes challenging, especially because it required balancing technical performance, cost, and practical implementation, but it was also very valuable and motivating.

I would like to sincerely thank my company promoter, Mr. Pedro Wyns, for his guidance, feedback, and practical suggestions throughout the project. I would also like to thank my college promoters for their support and advice during the development of this thesis. In addition, I would like to express my gratitude to everyone who supported me during this period.

Finally, I am grateful for the opportunity to work on this project, which allowed me to deepen my knowledge and gain practical experience in an area that strongly interests me.

Summary

HF radio stations employ several antennas since there is no perfect antenna for all operating frequencies and modes. Switching antennas manually is not convenient and involves a higher probability of making mistakes. The objective of the bachelor thesis was to design an intelligent remotely controlled antenna matrix which would allow switching between eight antennas and two HF transceivers, and which would include additional functionalities of measuring forward and reflected power, indicating SWR, and detecting operating frequency.

Starting from the system analysis and definition of functional, electrical, and mechanical requirements, a design was developed based on a two-board concept where the RF section was separated from control electronics. The RF board includes the antenna switching matrix, directional couplers, and frequency pickup circuits. The control board includes the CYD-based controller, relay drivers, ADC interface, and the control algorithm. The design includes relay selection, RF layout, grounding, and safety logic. Several revisions of relay selection were necessary due to high power stress in RF section under mismatch conditions.

As a result, an advanced design of HF antenna matrix was developed. It allows antenna switching, forward and reflected power measurement, SWR indication, and frequency detection. In addition, there is an opportunity to implement control functionality by means of remote communication. The project demonstrates that an intelligent antenna matrix can be realized using embedded design technology, considering proper relay selection and RF layout.

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Abbreviations

ADC — Analog-to-Digital Converter
APPCAD — Application Cad
CAT — Computer Aided Transceiver control
CYD — Cheap Yellow Display
DC — Direct Current
DPDT — Double Pole Double Throw
GPIO — General Purpose Input/Output
HF — High Frequency
I²C — Inter-Integrated Circuit
IC — Integrated Circuit
LCD — Liquid Crystal Display
PCB — Printed Circuit Board
RF — Radio Frequency
RMS — Root Mean Square
SCL — Serial Clock Line
SDA — Serial Data Line
SMD — Surface-Mount Device
SWR — Standing Wave Ratio
TX — Transmitter / Transceiver path
ULN — Darlington transistor driver array
VSWR — Voltage Standing Wave Ratio

Symbols

P — Power
P_f — Forward power
P_r — Reflected power
V_{rms} — Root-mean-square voltage
I_{rms} — Root-mean-square current
V_{peak} — Peak voltage
I_{peak} — Peak current
Γ — Reflection coefficient
Z_0 — Characteristic impedance
ε_r — Relative permittivity
λ — Wavelength
W— Microstrip width
H— Substrate height
T— Copper thickness

Start-up report

In the beginning of this bachelor thesis, the start-up report was completed and accepted by both parties. This document sets out the initial problem statement, the goals of the project, its deliverables, as well as preliminary planning of the project. The initial goal of the thesis was to make an intelligent remote controlled antenna matrix for shortwave radio which is able to switch up to eight antennas to two HF transceivers and provide local and remote control, forward and reflected power measurement, SWR warning, and failsafe behavior.

The start-up report also defines the intended deliverables of the thesis, among which are an RF switch board, a control system, a measurement part, a mechanical assembly, and all the corresponding documentation. Apart from it, an initial phased planning was made, which included phases such as analysis, system design, hardware design, firmware programming, prototype making, testing, integration, and writing the thesis.

However, in the process of developing the thesis, several issues concerning technical decisions required further analysis. These included relay selection, RF circuitry, grounding technique, and measurement approach. Therefore, the realization of the project was evolving compared to what was planned initially but it remained consistent with the primary tasks set out in the start-up report.

1. Introduction

Modern HF radio stations may contain more than one antenna in order to enhance the communication performance in different frequency bands and with different propagation conditions. Selection of an appropriate antenna for each case is needed for efficient operation, minimum reflected power and overall performance improvement. However, switching between multiple antennas in HF stations is often manual.

The aim of this bachelor thesis is to design and implement a remotely controlled intelligent antenna matrix for shortwave radio. The device is meant to switch eight antennas between two HF transceivers while guaranteeing safe operation at the RF levels. Besides switching itself, the device shall measure voltage standing wave ratio (VSWR), antenna power and optional frequency. Moreover, local status monitoring and remote-control functionality are desirable.

One of the biggest challenges in this project lies in combining all those functions together while considering the safety measures for such high-power RF devices. Switching circuits must reliably switch between antennas but should not cause unsafe relay states, wrong connections, improper antenna selection and switching when the transmitter is transmitting. Intended for remote control and monitoring usage, it should also provide watchdog and default state mechanisms as well as safe grounding of unconnected antennas.

Besides that, another critical issue of the project lies in RF signal handling and proper design of embedded circuitry to minimize electromagnetic interference and other effects caused by high RF level operation. It will be required to make proper decisions on separation between RF signal path components, embedded controllers and PCB layout in order to achieve desired results.

The above-mentioned tasks are typical for such a project, making it multi-disciplinary work. They include RF circuit design, measurement electronics, PCB design and embedded software design among others. Therefore, the main objective of this work will be development of a practical device for remote controlling HF radio antennas with additional monitoring capabilities.

This report provides a comprehensive overview of the entire design process from the initial idea and system requirements analysis to the design of PCB layout and assembly of the actual prototype with its evaluation and discussion.

2. Problem statement

The major aim of this project is to design an intelligent antenna matrix that would allow connecting eight HF antennas to two HF transceivers in a safe way. At an initial overview, this task looks like a pure problem of switching. However, it actually comprises several aspects related to high-frequency systems engineering: high power levels, reliable remote switching, low switching losses, and protection features, among others.

2.1. Operational need

Multiple antennas are used at HF radio stations since no single antenna is always good for all kinds of operations. The choice of the antenna is determined by the frequency range, propagation conditions, mode of operation, and the required radiation pattern. Switching of antennas is traditionally done by a human operator, and while it seems simple for one antenna and one radio transceiver, it becomes problematic for multiple antennas and two radios. For remote stations, the possibility to switch antennas manually becomes impossible, and, therefore, there is a necessity to design an automatically controllable antenna matrix.

Hence, the major requirement for the system is its ability to implement flexible antenna routing in a safe manner.

2.2. Main technical difficulty

Probably the most challenging part of the proposed system development is the high-frequency switching stage. According to the specifications, the solution shall support HF transceivers capable of transmitting power up to 1 kW. Such high power implies the use of specially designed high-frequency relays with contact capabilities both in terms of current capacity and reliability.
For a 50 Ω system at 1 kW, the matched RMS voltage and current are:

$$V_{rms} = \sqrt{P \cdot R} = \sqrt{1000 \cdot 50} \approx 223.6 \text{ V}$$
$$I_{rms} = \sqrt{\frac{P}{R}} = \sqrt{\frac{1000}{50}} \approx 4.47 \text{ A}$$

This already points out that the RF path must carry several amperes continuously. However, HF systems are not always perfectly matched. An important design consideration is the case of VSWR = 3:1.
For a VSWR of 3:1, the reflection coefficient magnitude is:

$$|\Gamma| = \frac{VSWR - 1}{VSWR + 1} = \frac{3 - 1}{3 + 1} = 0.5$$

The maximum standing-wave voltage and current are then multiplied by:

$$1 + |\Gamma| = 1.5$$

This leads to:

$$V_{max,rms} \approx 223.6 \times 1.5 \approx 335.4 \text{ V}$$
$$I_{max,rms} \approx 4.47 \times 1.5 \approx 6.71 \text{ A}$$

The corresponding peak values are:

$$V_{max,peak} \approx 335.4 \times \sqrt{2} \approx 474 \text{ V}$$
$$I_{max,peak} \approx 6.71 \times \sqrt{2} \approx 9.48 \text{ A}$$

These calculations show that the relay contacts and PCB traces may experience voltages and currents of approximately 335 V RMS, 474 V peak, 6.7 A RMS, and 9.5 A peak under mismatch conditions.
This makes relay selection one of the most difficult parts of the design.

2.2.1. Relay selection problem

In an ideal case, the technical solution should involve the usage of high-power coaxial RF relays like ***Figure 1***.

These relays are purposefully built for RF switching, with controlled impedance, better voltage spacing and power handling. Unfortunately, such relays are prohibitively expensive and would dominate the cost of the whole project. Thus, there was a need for a compromise between technical capabilities, availability, size and price.

As a matter of fact, during the course of the design, the relay selection problem occurred multiple times. This became one of the central challenges of the project.

Figure 1. High-power coaxial RF relay

A. Initial attempt: Omron G6K relay

Initially, the optimal solution seemed to be the usage of the Omron G6K relay seen in ***Figure 2***.
It is one of the smallest relays available, and hence it is attractive from a layout perspective. Using G6K, it would be possible to create a compact board with 8×2 switching matrix.

However, after further consideration of this relay, it turned out that it belongs to small-signal relays and thus is not suitable for high power applications. Its current and voltage ratings, as well as its contact spacing and mechanical robustness were insufficient for the purposes of this project. Despite the usage of both poles in parallel, the relay remained insufficiently powerful.

Thus, a very compact relay seems to be very attractive, but it may not be enough to build a reliable 1 kW HF RF switch.

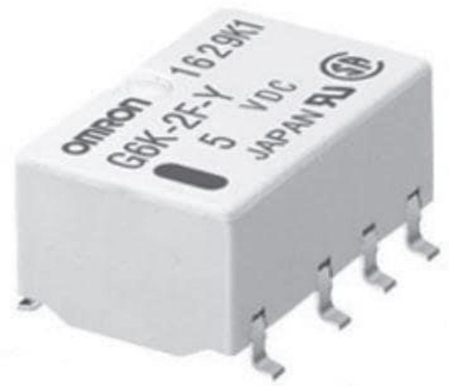

Figure 2. Omron G6K-2f-Y Relay

B. Next stage: Search for an alternative

After excluding the G6K as the relay for the RF circuit, the next task to address became whether there exists a larger but still relatively cheap relay with significantly better current ratings and voltage tolerance.

At that stage, one of the main comments from the supervisor was that each contact must be capable of carrying at least 3 A current. In particular, this was due to the fact that the design uses DPDT relays with both poles working in parallel. Having at least 3 A per pole makes the current sharing more realistic.

i. Larger relay: Omron G2RL

One of the next relays to consider was the Omron G2RL family (***Figure 3)***.
It was better suited in terms of current rating and mechanical robustness compared to the G6K relay. It appears to be a true power relay and provides much greater confidence in regard to its ability to function under high power.

Unfortunately, it also has one serious disadvantage – size. It is significantly larger than G6K relay and thus makes the layout of 8 channels much more complicated. This creates additional pressure on PCB size and routing.

Figure 3. Omron G2RL Relay

ii. SMD Relay alternative: IM06DGR

As the need for compactness of the board did not disappear and an SMD solution looked promising, there was a need to find a proper relay of the same family. One of the options was the family of TE Axicom IM series relays. In particular, the IM06DGR model (***Figure 4)*** is a DPDT, 12V and 5A SMD relay.

This relay is a good replacement for the G6K as it has much more power capacity compared to the initial design solution. At the same time, it is smaller than G2RL and thus offers advantages in terms of size.

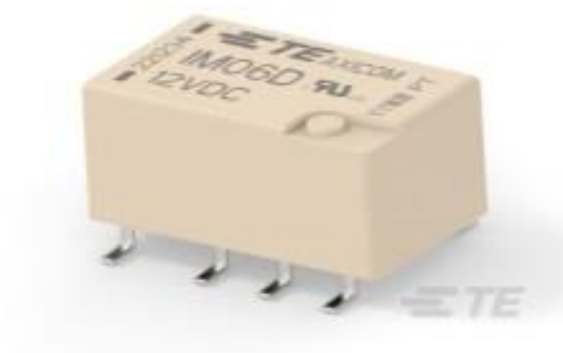

Figure 4. TE Axicom IM06DGR Relay

However, similar to other relays of this family, IM06DGR is still not optimized for RF operation. Thus, it cannot be installed arbitrarily and requires rules such as:

- Both poles in parallel
- No hot switching
- Short RF paths
- Good 50Ω bus layout
- Software interlocks to prevent entering an invalid state.

Thus, regarding relay selection there appeared a clear need for trade-offs between RF capabilities, current ratings, package size, mechanical strength, PCB space and price.

2.2.2. Ground-clamp relay problem

Apart from the mentioned problems, another challenge in the relay matrix design involved implementing the ground-clamp relays for the unused antennas. The function of the ground clamp is to make sure that any antenna not being used is electrically grounded, since leaving antennas floating creates potential coupling issues and adds unpredictability and unreliability to the overall system behavior.

At the beginning of the design process, it was realized that there was a mistake in the way the wiring of the clamp relay is organized. Namely, the ground has been placed to the wrong contact side, which means that the antenna will remain grounded in its default position. As a result, special attention was paid to understanding how the COM, NC, and NO contacts in a typical relay work. The right solution was found: the antenna node must be wired to the common contacts, while the ground – to the normally-open contacts, so the antenna is grounded only in the case when the clamp relay is closed on purpose.

- **Use of the same relay type for ground clamp**

One of the other decisions during the design process was to choose the same relay family for both the switching relays and the ground clamp relays. In particular, at first sight it looked appealing to use a smaller relay as the clamp element. The reasoning behind such a decision was the idea of saving PCB area. However, since the clamp relay is directly connected to the antenna RF node as well, the clamp relay may potentially be subjected to RF-related voltage and current stress. Using a weak relay as the clamp relay would create an additional weak point in the RF chain, and, therefore, a stronger relay should be used instead.

2.3. Measurements and integration difficulties

As stated above, the project included the implementation of several additional functions apart from the switching matrix functionality. In addition to switching, the antenna matrix must also support power measurement, frequency detection, safety logic, and remote control.

2.3.1. Forward and reflected power measurements and SWR calculation

An important requirement of the proposed RF system is the ability to measure both the forward power and the reflected power of the transceiver output in order to detect mismatch conditions. From these two quantities, the Standing Wave Ratio (SWR) can be determined, which provides an indication of how well the antenna system is matched to the transmitter.
The Standing Wave Ratio can be calculated from the measured forward power P_f and reflected power P_r using:

$$SWR = \frac{1 + \sqrt{P_r / P_f}}{1 - \sqrt{P_r / P_f}}$$

Figure 5. SWR calculation formula.

where P_f is the forward power and P_r is the reflected power. This relation is based on the reflection coefficient, which can be expressed in terms of the ratio between reflected and forward power. When the reflected power is very small compared to the forward power, the antenna system is well matched and the SWR approaches 1:1. As the reflected power increases, the SWR also increases, indicating a poorer impedance match between the transmitter and the antenna.

This formula is particularly useful in the proposed system because the forward and reflected outputs of the directional coupler can be measured separately. Once these two quantities are available, the controller can estimate the SWR in software and use it for monitoring, warning, or protection purposes.
For this purpose, two possible directional coupler topologies were considered: the Bruene coupler and the tandem coupler. Although the tandem coupler is known for its good directivity, it is more complex to design and manufacture. The Bruene coupler was considered a more practical solution for the present prototype because it combines a voltage sample and a current sample to separate the forward and reflected components of the RF signal. In addition, it is a well-established approach in HF wattmeter and SWR meter designs.

For these reasons, the Bruene coupler was selected for this project. It provides the required measurement principle while being simpler to implement and requiring fewer components than the tandem coupler. Furthermore, the use of adjustable trimmer capacitors makes it possible to balance the bridge during testing, which improves directivity and increases the accuracy of the reflected-power measurement.

2.3.2. Frequency measurement

One more desirable functionality of the system is the ability to measure the transmitted RF frequency. At the initial design stage, CAT control was proposed as one of the alternatives because it allowed reading the operating frequency from the transceiver directly. However, this option required the presence of a proper communication interface and particular protocol of operation of the radio device. Therefore, this method could hardly be considered universal enough for a prototype designed to be as independent from the exact transceiver model as possible.

For this reason, it was decided to follow the hardware implementation of the RF frequency measurement. This functionality is provided by the combination of an RF pickup capacitor, an attenuation and protection stage, and the LMV7219 comparator. The RF pickup capacitor takes the sample of the RF signal sent from the transceiver, and then this signal is attenuated and protected in order to supply it to the comparator, where it is transformed into a pulse stream of digital values readable by the controller.

On the one hand, the principle behind the measurement process seems rather simple, but, on the other hand, implementing this functionality is somewhat problematic. In particular, due to the limited availability of free GPIO pins of the selected CYD controller platform, there is little choice when it comes to implementing the frequency measurement functionality with two inputs.

2.4. System Integration

The matrix cannot be treated as a stand-alone RF switch, because it must also integrate measurement, control, remote communication, and safety logic. As a result, the project must be considered as a complete RF and embedded control system rather than only a relay-based switching network.

2.5. Conclusion of the problem analysis

Therefore, the problem tackled in this thesis does not just relate to switching antennas; it relates to the ability to do so safely, efficiently, and economically while under the high-power conditions characteristic of HF. One of the main challenges encountered during the course of the project concerned the selection of the relay itself, due to the often-competing factors of power capacity, size, cost, manufacturing ability, and PCB space.

Moreover, there was the added challenge of integrating RF switching with power measurement, SWR measurement, frequency measurement, embedded processing, and remote control. Each of these systems presented additional design considerations and, when combined, added greatly to the complexity of the problem to solve a simple RF switch.

Therefore, the numerous design changes that took place throughout the duration of this project were not necessarily an indicator of poor planning, but instead resulted from increasingly accurate calculations, guidance by the thesis advisor, and a greater understanding of the RF and system limitations.

3. Method

The development of the system was carried out in phases, starting with analysis of the assignment and continuing through practical implementation and testing.

3.1. Analysis of the assignment and definition of requirements

During this step, the description of the assignment was analyzed and converted into a set of requirements. These included switching eight antennas between two HF transceivers, providing local and remote control, measuring forward and reflected power, monitoring SWR, optionally measuring frequency, and implementing fail-safe and watchdog functions.

The main limitations were stated during this step. These were related to functionality, electrical parameters and dimensions of the system. Specific attention was drawn to the level of RF power, reliable switching and RF/Digital separation.

3.2. Literature study and technical comparison

After defining the problem and its requirements, technical comparisons and literature review were carried out. This included several approaches to solving critical issues within the design. They included the following items:

- Type selection for relays used in RF switch matrix;
- Grounding concepts for unused antennas;
- Variants of directional coupler implementation for measuring power and SWR;
- Frequency detection techniques.

For the subsystem of power measurement, two couplers could be chosen – the Bruene coupler and the tandem coupler. After comparing both, the Bruene coupler was selected as a more compact variant which can be implemented on a PCB with some tuning capabilities.

When designing the system for frequency measurement, CAT control interface was first proposed. However, later this approach was rejected as an RF detector-based solution was found more appropriate. It includes RF pickup, attenuation, signal protection and pulse generator based on comparator IC.

3.3. Engineering calculations and design validation

In order to support component selection and validate the switching concept, the RF voltage and current levels in the circuit were calculated. Calculations were performed assuming 50 Ω system and 1 kW of design power.

Moreover, a mismatch calculation at the point of VSWR = 3:1 was performed to estimate maximum voltage and current levels which might occur when switching relays, traces on PCB and measurements circuitry.

These results were essential for relay and PCB design.

3.4. Schematic design

Before developing complete schematics, the system was first represented as a functional block diagram. As shown in ***Figure 6.***, the design is divided into an RF section and a control section, with additional blocks for power measurement, frequency detection, relay driving, and user control.

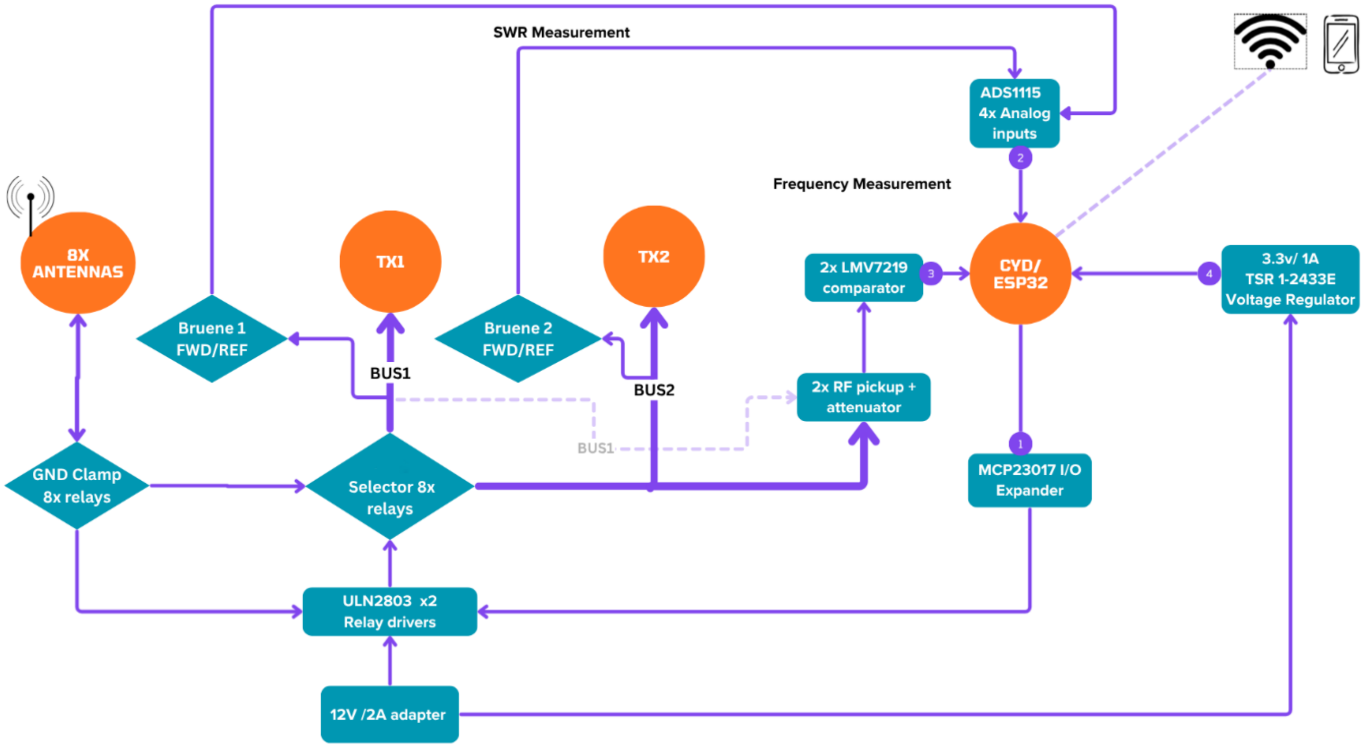

Figure 6. High-level block diagram of the proposed antenna matrix system.

The above block diagram explains how the signal flow and interaction among the different systems occurs. The signals from the antenna pass through the selector/ground clamp relay circuitry to connect to one of the two RF buses feeding the transceivers. There is a Bruene Coupler within each of these RF paths that is used to measure the reflected and forward powers, along with the RF pick-up and comparator that is required for detecting frequencies. The analog and digital signals generated are then passed on to the control board.

3.5. PCB design

The PCB design was implemented as two separate boards: an RF board and a control board. Particular attention was given to short RF paths, relay orientation, controlled grounding, and the separation of RF and digital circuitry.

3.6. Component selection and ordering

After completing schematics and PCB design, the next step was components selection and ordering. This step also included trade-offs between performance, package type, availability, and cost. Final choices were based not only on the data sheets, but also on practical considerations of PCB area and ease of assembly.

3.7. Assembly and testing

The prototype was assembled step by step, beginning with the most essential hardware blocks. Testing was planned progressively so that individual subsystems could be checked before moving to full system integration.

3.8. Software development and system integration

The software development included relay control, safety logic, measurement data processing, and user interaction functionalities. For this purpose, CYD platform was used as a control interface. The limited number of easily accessible GPIO pins on the CYD introduced additional constraints during software integration.
Software development was performed along with hardware testing, allowing to validate control logic and measurement circuits step by step during assembly.

3.8.1. Approach to software development

Software is developed gradually, starting with the simplest possible functions: communication with MCP23017, basic relay output tests, ADC reading, and frequency detection via GPIO pins. Once these tasks are accomplished, more advanced functionality may be developed.

3.9. Documentation

Throughout the project, the thesis documentation was developed in parallel with the hardware and software realization. This included recording of design decisions, calculations, changes in schematics and PCBs, and testing steps. This way, documentation reflects not only the final design, but also the process of engineering.

3.10. Conclusion of the methodology

It should thus be noted that the methodology of this thesis project can be described as an iterative engineering process. The project did not follow a linear path due to the need to revise some design choices. However, such iterative approach was necessary and valuable, allowing to improve the design progressively based on analysis, calculations, feedback, and practical issues.

4. Research

4.1. System requirements

The system requirements are derived based on the project description, the application purpose of the antenna matrix, and the technical limitations established in the analysis phase. Requirements outline the operational and technical capabilities required from the final prototype of the device.

These include:

1. functional requirements;
2. electrical and RF requirements;
3. safety and reliability requirements;
4. mechanical and integration requirements.
5. power supply requirements
6. implementation requirements

4.1.1. Functional requirements:

The system shall provide the capability to switch several antennas between two HF transceivers. Functional requirements include the following:

- The system shall support connection of 8 antenna inputs.
- The system shall support two HF transceiver outputs.
- Each antenna input shall be able to connect to either of the two transceiver outputs.
- Unused antennas shall be placed into a safe state through the implementation of a ground-clamp function.
- The system shall feature a local monitoring and control functionality with a CYD display.
- The system shall be remotely controllable from the Android smartphone application via a Blynk interface.
- The system shall include the functionality of measurement of forward and reflected powers for each path separately.
- The system shall estimate the standing wave ratio (SWR) from the measured forward and reflected power.
- It is desirable that the system be capable of measuring the transmitted RF frequency on both transceiver paths.
- The system shall incorporate logic to avoid invalid or unsafe switching states.

4.1.2. Electrical and RF requirements

As the system is expected to operate in HF transmission regime at relatively high power, electrical and RF specifications play a particularly important role.

- The system shall be designed to work in a 50 Ω RF environment.
- The system shall be able to accommodate a design power level up to 1 kW of continuous RF power.
- The RF paths shall be designed to minimize the insertion loss of the switching function.
- The RF design shall be optimized to ensure minimal RF mismatches and reflections.

- Main RF paths shall be implemented via short-trace, low-impedance tracks; the appropriate 50 Ω microstrip bus design shall be utilized where appropriate.
- RF circuits shall be arranged to minimize the cross-coupling between the RF lines and ensure reliable and stable operation of measurement circuits.
- The directional coupler subsystem shall provide separate forward and reflected measurement signals for SWR estimation.
- Power and measurement circuitry shall be designed with sufficient voltage and current margins with respect to expected mismatch conditions.

4.1.3. Safety and reliability requirements

Dealing with a remote controlled system and high RF power creates the following requirements:

- The system shall operate safely and reliably under normal and fault conditions.
- The system shall implement an SWR warning or protection mechanism.
- The system shall prevent from establishing an unsafe relay combination by incorporating the interlocking logic.
- The system shall prohibit or block switching of the relays when active RF transmission occurs.
- In case of failure of the system power supply, automatic switching to an emergency mode which operates with the main antenna should be performed.
- Unused antennas should not remain open but be placed into a controlled state.
- The watchdog functionality shall be implemented to ensure higher reliability of the system when operated remotely.
- The design shall minimize the risk of dangerous switching states caused by operator error.

4.1.4. Mechanical and integration requirements

There are restrictions set by the assignment which should be accounted for in the mechanical aspects of the design.

- The system shall be designed in the form of two separate PCBs.
- The RF and control boards shall be physically separated and laid out to minimize RF interference.
- The system enclosure shall be manufactured using die-cast aluminium enclosure(s).
- The system design shall allow integration of multiple RF connectors and required internal interconnections.
- The layout shall enable assembly, testing, tuning and debugging of the design.

4.1.5. Power supply requirements

The prototype needs two different power sources: one for the relay, which uses a higher voltage, and another for the logic, which uses a lower voltage.

- The system shall operate from an external 12 V power supply.
- The power source for the relay coils should be connected to the right-hand high-side supply line.
- The controller and low-voltage electronics shall operate from a regulated 3.3 V supply.
- The power distribution shall be designed to avoid interference between relay currents, digital electronics, and sensitive analog measurement circuits.

4.1.6. Implementation requirements

The project includes the following requirements and rules, additionally to its basic functionalities:

- The design shall remain economically realistic for a prototype.
- Component choices shall balance cost, availability, size, and performance.
- The design shall be manufacturable using available PCB technology and accessible assembly methods.
- The prototype shall be testable in stages, allowing validation of relay control, measurement circuits, and software separately before full system integration.

4.2. **Hardware design**

The hardware of the proposed antenna matrix was designed to combine high-power HF switching, RF measurement, digital control, and safe system operation in a single prototype. Due to the combination of RF and digital parts, the hardware was divided into two boards – the RF board and the control board. Such separation reduces interference between RF and digital parts and meets the project requirement according to which the RF part should be well-shielded from the microcontroller part.

4.2.1. Overall hardware architecture

The complete hardware system consists of the following building blocks:

- Antenna switching matrix,
- Ground-clamp relays for unused antennas,
- Directional couplers for forward and reflected power measurement,
- Frequency pickup and conditioning circuits,
- Relay driver circuitry,
- Analog measurement interface,
- CYD-based control platform,
- And power supply and regulation.

The RF board contains the antenna switching matrix, the directional couplers, and the RF frequency pickup circuits. The control board contains the CYD controller, relay driver electronics, analog measurement interface, and supporting circuitry.

4.2.2. RF switching matrix

The main function of the hardware is a switching matrix that connects 8 antenna inputs to 2 HF transceiver outputs. Each antenna path contains the following relays:

A. Selector relay – to connect the antenna to BUS1 or BUS2;
B. Ground-clamp relay – to ground the antenna if it is not selected.

Both relay functions are illustrated in ***Figure 7.***

This configuration allows each antenna to be placed in one of the three valid states:

- Connected to transceiver Bus 1,
- Connected to transceiver Bus 2,
- Grounded in the parked state.

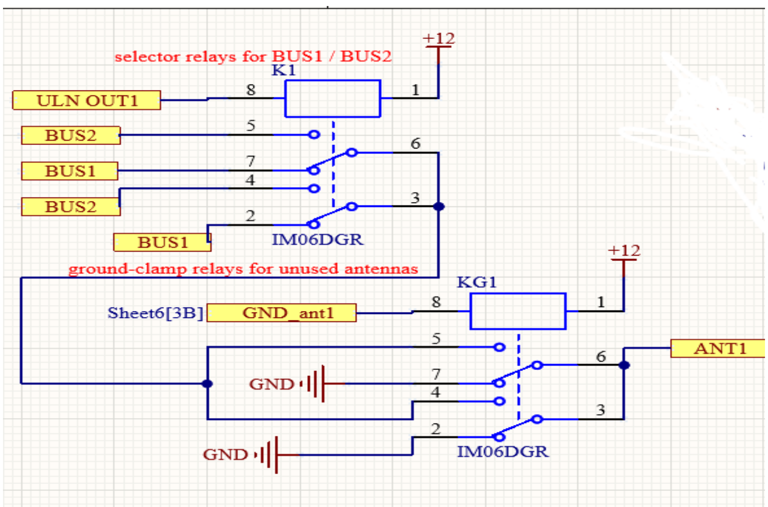

Figure 7. Selector relay and ground-clamp relay schematic

A. **Selector relays**

The RF switching matrix uses DPDT relays with both poles connected in parallel in order to reduce contact resistance and share current. The final relay choice was based on the current and voltage requirements discussed in the problem analysis.

B. **Ground-clamp relays**

As unused antennas should not be left floating, each antenna is equipped with a ground-clamp relay. Its purpose is to reliably ground the unused antenna.

Clamp relays are also implemented with both poles in parallel, since they are directly connected to the antenna RF node and thus belong to the RF power path. Though it seems optimal to use a smaller relay for the clamp function, such implementation would introduce a weak link into the switching network. Therefore, the same relay family was selected for both functions.

4.2.3. RF buses and PCB layout strategy

The RF board was designed around two main RF buses:

- BUS1
- BUS2

These buses represent the two transceiver paths through the switching matrix. In order to improve the RF behavior of the design, the layout was developed around short microstrip traces with a solid ground reference underneath. This approach was selected to obtain a more controlled impedance and to reduce unwanted reflections and parasitic effects in the RF path.

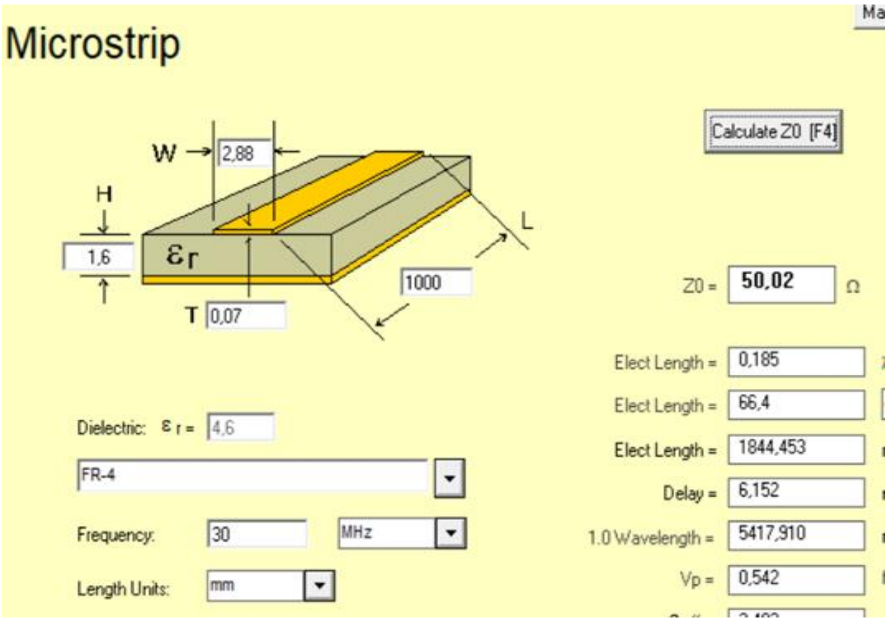

Figure 8. AppCAD microstrip calculation used for the 50 Ω RF bus design.

The width of the main RF traces was calculated using the AppCAD microstrip calculator. Based on the selected PCB parameters, including substrate thickness, dielectric constant, and copper thickness, the trace width was chosen so that the characteristic impedance of the bus is approximately 50 Ω. This is important because the antenna matrix is intended to operate in a 50 Ω RF system and maintaining this impedance as closely as possible helps reduce mismatch and insertion loss.

Specific attention was given to:

- Relay orientation,
- Short RF trace lengths,
- Sufficiently wide copper traces,
- Minimization of parasitic bus length,
- Strong grounding.

The RF traces were designed to be as straight and short as possible between the antenna nodes, relay contacts, buses, and transceiver outputs. In addition, the ground-clamp paths were also kept short and low-impedance in order to ensure an effective grounding function for unused antennas.

4.2.4. RF connectors

For the connections to the external RF circuits, the chosen connectors are N-type connectors for the antenna connections as well as for the transceiver connections. This choice was made because the system is intended to operate at relatively high power in the HF range, where mechanical robustness and reliable RF connection are essential.

The N-type connectors seem to be preferable over the SMB connectors for several reasons. First, they offer significantly improved mechanical strength. Secondly, they are better designed for use at high RF power, whereas SMB connectors are typically used in signal-level or compact RF systems.

Thirdly, the N-type connectors are frequently used in practice and are more suitable for RF systems installed in a shielded metal enclosure.

Figure 9. N-type connectors

4.2.5. Power and SWR measurement

To monitor the RF performance of the system, two directional measurement circuits are included, one for each transceiver path. These circuits provide information about:

- Forward power,
- Reflected power,
- The standing wave ratio (SWR).

The selected measurement principle is based on the Bruene coupler. This coupler was chosen because it provides a practical and relatively simple solution for separating the forward and reflected components of the RF signal in an HF system. It is therefore well suited to the requirements of this prototype.

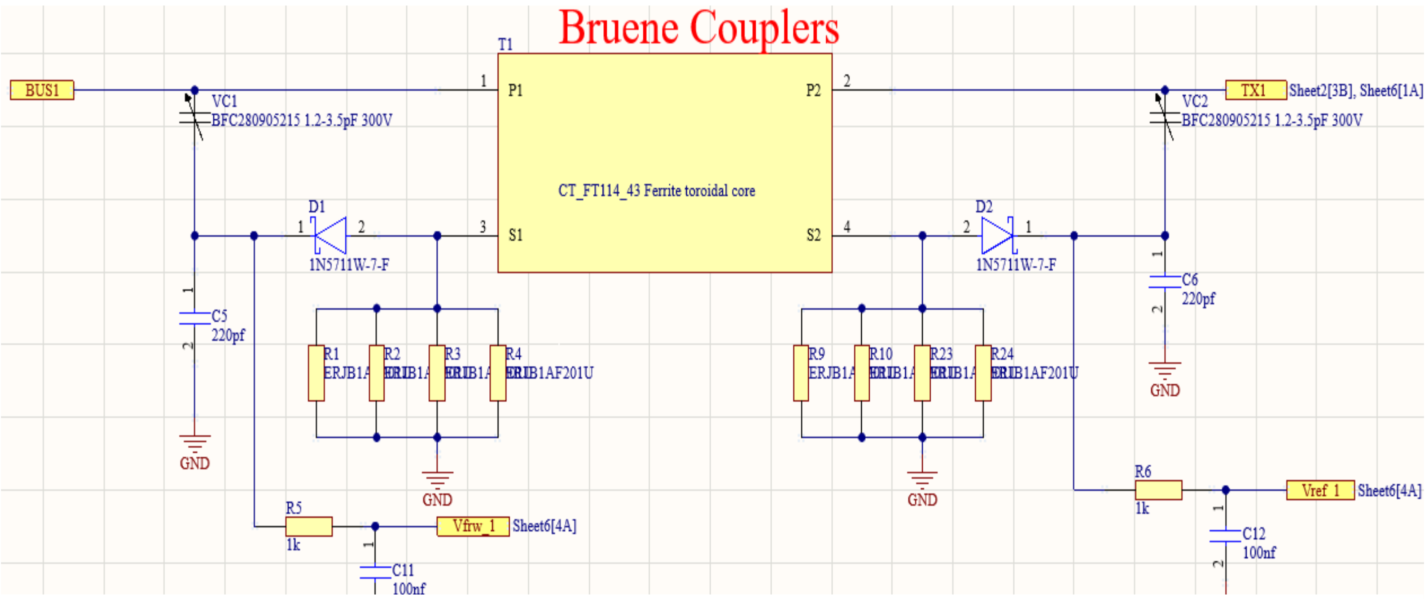

Figure 10. Bruene coupler schematic

The Bruene coupler shown in *Figure 10.* consists of:

- A current transformer built around a ferrite toroidal core,
- A voltage sampling network,
- Bridge balancing components,
- Rectifying diodes,
- Filtering circuitry.

The small bridge sampling capacitors are implemented as adjustable trimmers so that the coupler can be tuned for improved balance and directivity during testing. In addition, the bridge load resistors are implemented as parallel resistor networks in order to obtain the required equivalent resistance with sufficient total power dissipation capability.
The outputs of the couplers provide low-frequency analog voltages proportional to the forward and reflected RF power. These analog signals are then transferred to the control board, where they are digitized and further processed by the controller to estimate power levels and SWR

4.2.6. Frequency measurement hardware

In addition to power measurement, the hardware also includes a frequency detection function. A hardware-based solution was selected in order to keep the system independent of transceiver-specific communication protocols.

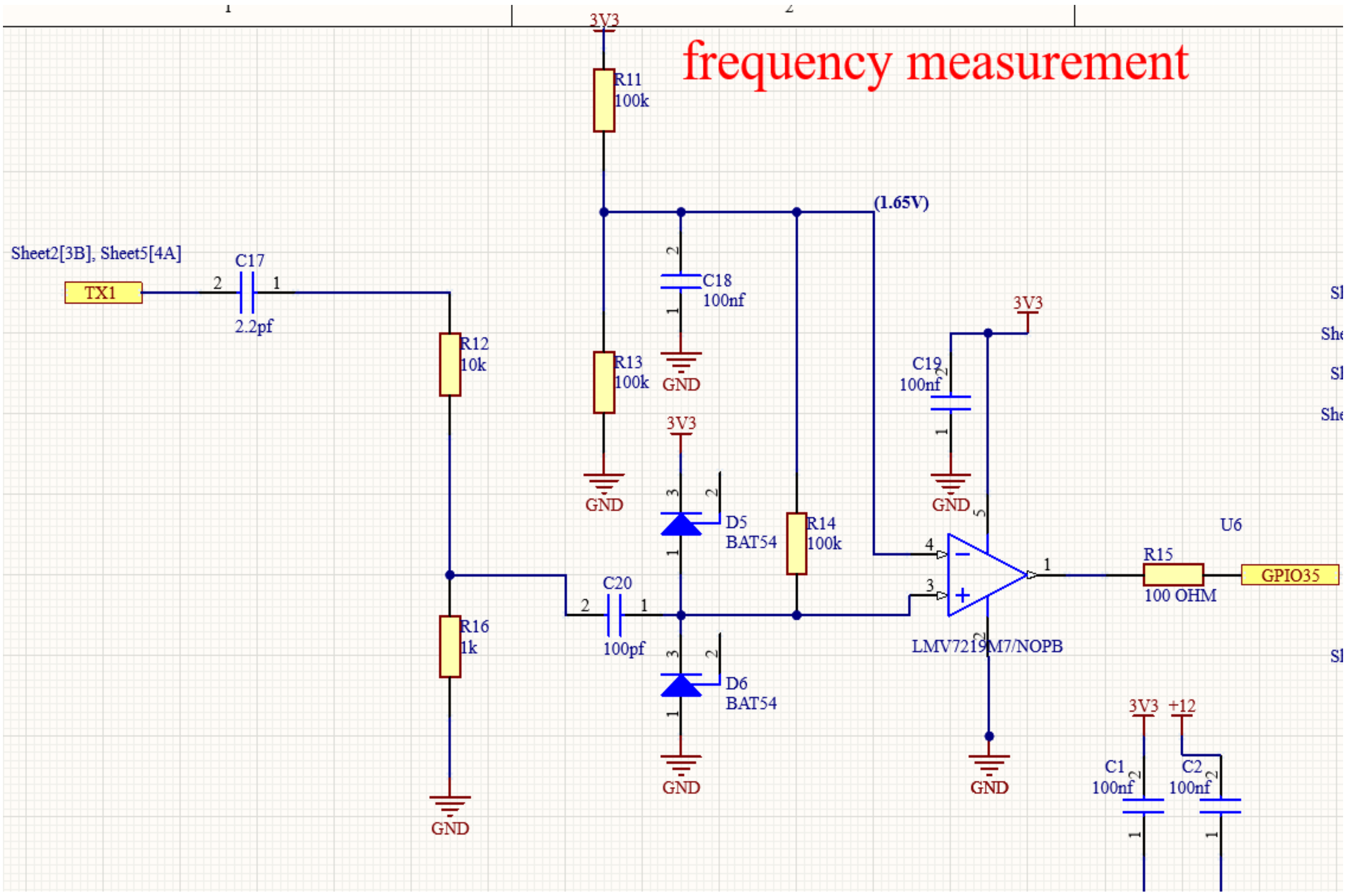

Figure 11. Frequency measurement system schematic

Each transceiver path shown in ***Figure 11***. includes:

- A small RF pickup capacitor ***(C17),***
- An attenuation and protection stage (***R12,R16 and C20)***
- An LMV7219 comparator.

The RF pickup capacitor samples only a very small fraction of the transmitted RF signal. This sampled signal is then attenuated and protected before being applied to the comparator input. The comparator converts the analog RF waveform into a clean digital pulse signal, which can then be counted by the controller.

The implementation of this function is also influenced by the limited number of easily accessible input pins on the CYD platform, which creates an additional integration constraint.

4.2.7. Control board

The control board contains the digital and low-voltage hardware required to operate the system. The main elements are:

A. CYD platform with ESP32,
B. MCP23017 I/O expander,
C. ULN2803 relay driver stages,
D. ADS1115 analog input interface,
E. The supply distribution circuitry.

A. CYD platform

The CYD is used as the main controller and local display platform. It provides the processing capability for relay control, display handling, remote communication, and measurement processing. It also serves as the software platform for the user interface and Blynk communication.
A limitation of the CYD is the relatively small number of easily accessible external I/O pins. This required careful pin allocation and the use of peripheral expansion devices.

Figure 12. CYD board

B. MCP23017 I/O expander

Because the number of relay control signals exceeds the directly available GPIO resources, the design uses the MCP23017 I2C I/O expander. This provides additional digital outputs for relay control and allows the CYD to manage a large number of relays through only a small number of control lines.

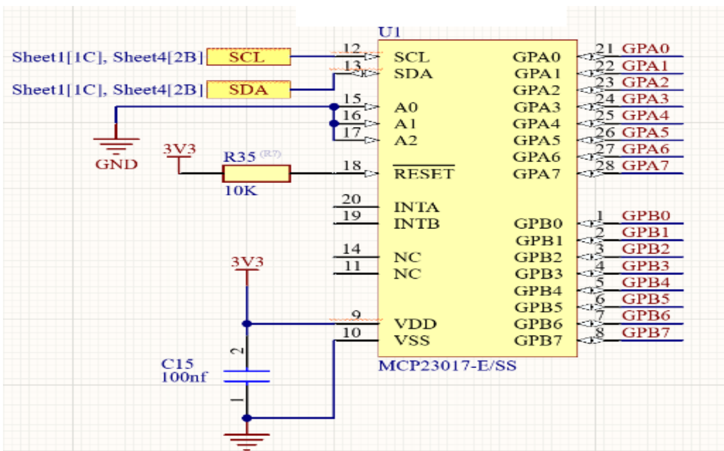

Figure 13. MCP23017 I/O expander

C. ULN2803 relay drivers

Relay coils cannot be powered directly by the microcontroller or by the MCP23017 I/O expander since such devices are designed only for use with low-power digital logic signals. Relay coils consume relatively high current in comparison to what can be provided by the microcontroller GPIO pins or the I/O expander pins. In addition, relay coils require 12 V, while the CYD, as well as the MCP23017, work using the 3.3 V logic levels. Thus, some interface stage should be implemented between the low-voltage control electronics and the relay coils.

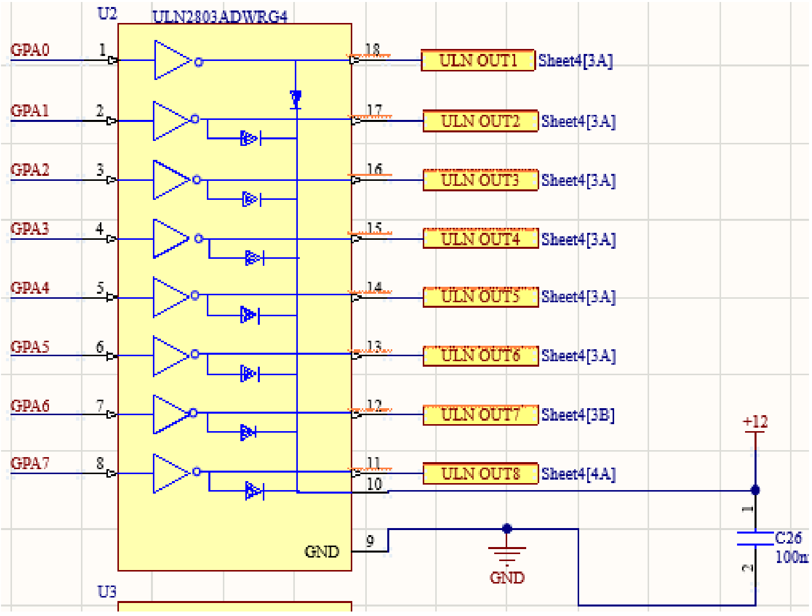

Figure 14. ULN2803 relay drivers

To implement the required functionality, ULN2803 Darlington driver arrays are used as an interface between MCP23017 outputs and relay coils. The ULN2803 performs the role of current amplifier: it accepts the low current logic signal as input and switches a relatively high current required by the relay coil. This allows the microcontroller and I/O expander not to power the relay coils directly.

Another function performed by the ULN2803 is protection. Relays are inductive loads and produce a significant voltage spike at the time of turning off. The ULN2803 incorporates an internal flyback diode which protects the rest of the circuitry from these switching transients.

Thus, the ULN2803 performs three important tasks in this implementation:

- It provides current amplification for relay coils operation,
- It provides level interfacing between the 3.3 V control logic and the 12 V relay supply by acting as a low-side driver.
- It provides protection against inductive switching transients.

D. ADS1115 analog interface

The analog outputs of the Bruene couplers cannot be processed directly in their analog form by the control logic. For this reason, an ADS1115 analog-to-digital converter (ADC) is used to convert the measured analog voltages into digital values that can be read by the controller.

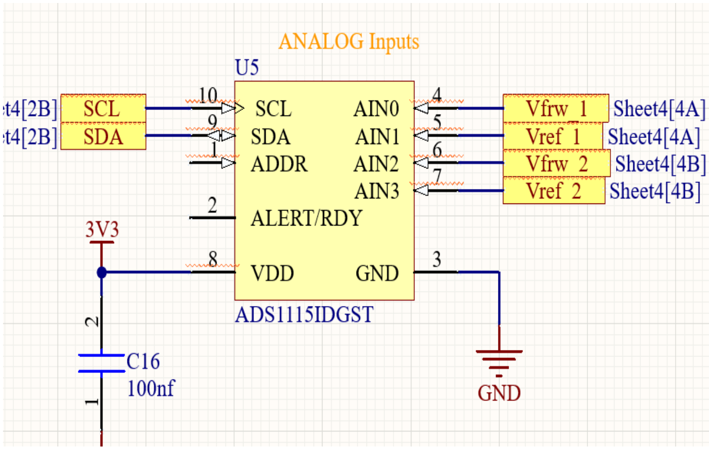

Figure 15. ADS1115 analog interface

The ADS1115 was selected because it provides four analog input channels, which is exactly sufficient for the four required measurement signals in this project:

- Forward power for transceiver path 1,
- Reflected power for transceiver path 1,
- Forward power for transceiver path 2,
- Reflected power for transceiver path 2.

In addition, the ADS1115 communicates through the I2C interface, which makes it easy to integrate with the CYD-based controller while using only a small number of communication lines. This is particularly useful in the present design, where the available GPIO resources are limited.

The Bruene couplers provide low-frequency analog voltages proportional to the detected forward and reflected RF power. These signals are connected to the analog input channels of the ADS1115, where they are digitized and then transferred to the controller. The controller can then use these digitized values to estimate power levels and calculate the SWR in software.

As a result, the ADS1115 provides a compact and practical interface between the analog power measurement circuitry and the digital control system.

E. Power supply hardware

The system is powered from an external 12 V DC supply. This supply is used directly for the relay section, because the relay coils require a higher operating voltage for reliable switching. However, the control electronics, including the CYD, the MCP23017, the ADS1115, and the comparator-related logic, require a regulated 3.3 V supply. For this reason, the power architecture is divided into two voltage domains: a 12 V relay supply and a 3.3 V logic supply.
To generate the logic supply, a TSR-1-2433E step-down regulator is used.

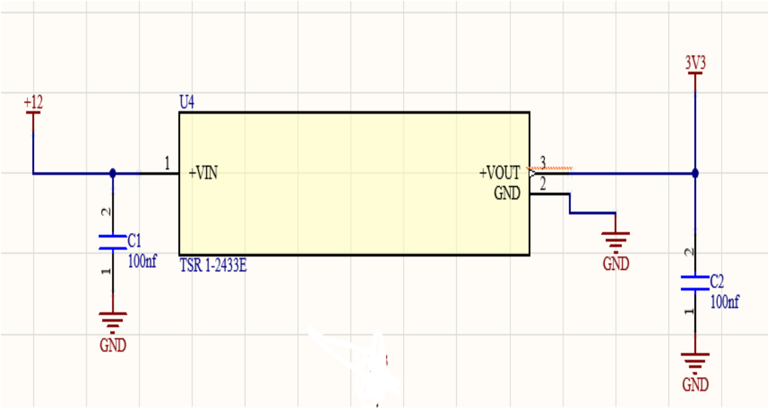

Figure 16. TSR-1-2433E 3.3 V step-down regulator

This regulator converts the 12 V input voltage into a stable 3.3 V output suitable for the low-voltage electronics. The use of a switching regulator is advantageous because it provides good efficiency and avoids the higher power dissipation that would occur with a linear regulator when converting from 12 V to 3.3 V.
The power architecture therefore includes:

- A main 12 V input supply,
- A regulated 3.3 V rail for the CYD and other low-voltage electronics,
- Distribution of the 12 V relay supply to the driver stages.

A local 100 nF decoupling capacitor is placed at both the input and output of the regulator in order to improve supply stability and reduce high-frequency noise. In addition, a dedicated 12 V power connector is used to bring the external supply into the system.

Relay current paths were separated from sensitive analog and digital sections, especially on the control board. This is important because the relay coils draw switching current and can introduce noise into the supply network. By separating these current paths and keeping the logic supply properly regulated, the stability of the control and measurement circuits is improved.

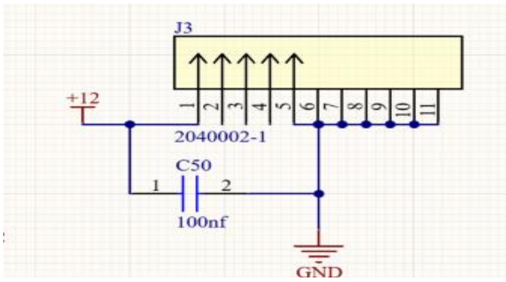

Figure 17. 12 V power input connector.

4.2.8. Board separation and shielding

Because the project combines high-power RF switching and digital control electronics, board separation is an important part of the hardware design. The RF board and the control board are separated both electrically and physically. This reduces RF coupling into the controller section, improves measurement stability, and makes shielding easier to implement.

The RF board was designed as a 4-layer PCB. This choice was made to obtain a better RF structure than would be possible with a simple 2-layer board. On the RF board, the top layer is used for the main RF traces, while the second layer is a continuous ground plane directly underneath. This provides a stable reference plane for the RF paths and allows the implementation of controlled microstrip routing. The third layer is used as the +12 V power plane, which supports relay power distribution while keeping the supply routing organized and separated from the main RF traces. The bottom layer is used for additional routing and support connections.

This 4-layer structure improves the design in several ways. First, the short distance between the top RF traces and the internal ground plane makes the RF return path more controlled. Second, it reduces unwanted coupling and improves shielding between the RF section and the rest of the hardware. Third, it allows the RF traces to be designed more predictably as microstrips than on a standard 2-layer board.

The RF board therefore focuses on:

- Short RF paths,
- Controlled grounding,
- Microstrip routing,
- Coupler integration,
- A solid internal ground reference.

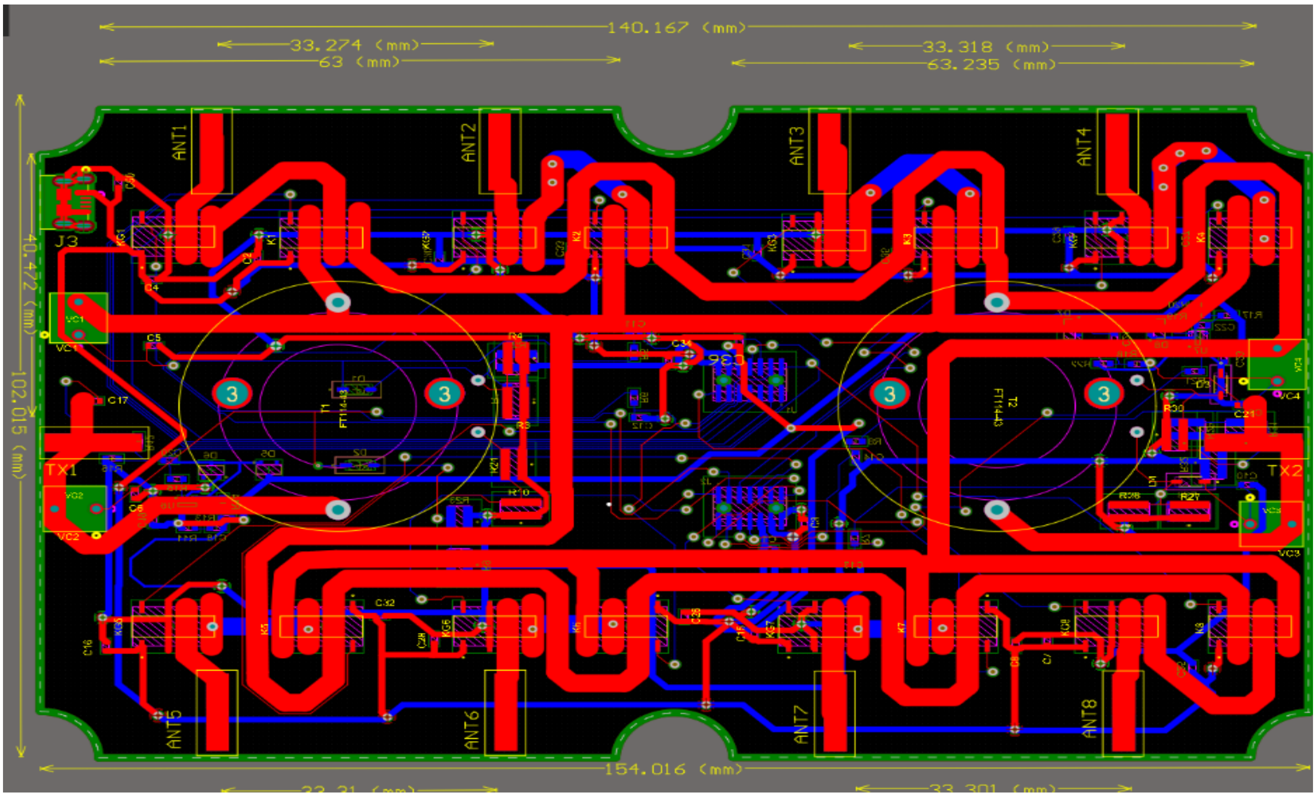

Figure 18. RF board layout

The control board was implemented as a 2-layer PCB. On this board, the top layer is mainly used for routing the control and interface signals, while the bottom layer is kept as a ground plane by means of a continuous copper pour as much as possible. This provides a solid ground reference for the digital and analog control circuits, improves return-current paths, and helps reduce noise coupling on the board.

The control board focuses on:

- Logic,
- Relay driving,
- Analog acquisition,
- User interface support,
- A grounded bottom layer for improved noise control.

By separating the RF hardware from the digital control electronics, the design becomes easier to route, easier to debug, and more suitable for shielding inside the enclosure. This separation also helps to reduce the influence of relay switching and digital control activity on the sensitive RF and measurement circuitry.

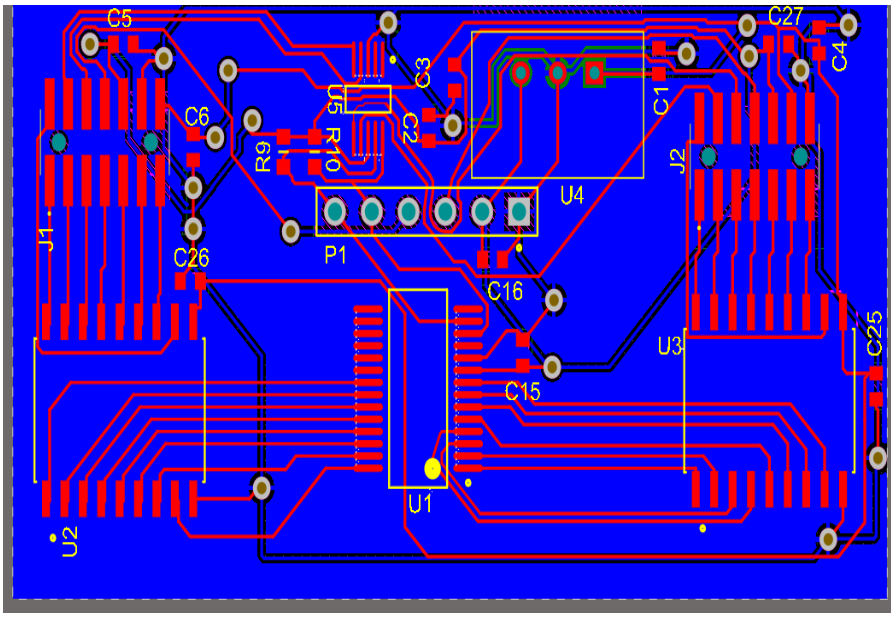

Figure 19. Control board layout

4.2.9. Enclosure and mechanical integration

For the enclosure, a Hammond 1550WEBK die-cast aluminum housing is used (***Figure 20***). This type of enclosure was selected because the project combines high-power RF circuitry and digital control electronics, which makes mechanical robustness and electromagnetic shielding important design considerations.

A metal enclosure is preferred because it provides much better shielding than plastic housing. This helps reduce unwanted RF coupling, limits external interference, and improves the separation between the RF section and the control electronics. In addition, a die-cast aluminum enclosure is mechanically strong and well suited for mounting RF connectors and supporting the complete prototype in a practical way.

The enclosure will be machined to create the required openings for the N-type RF connectors.

Figure 20. Hammond 1550WEBK aluminum housing

These openings are planned to be made using a CNC machine, which allows accurate positioning and repeatable hole placement on the enclosure walls. This is important because the RF connectors must be mounted with sufficient mechanical precision and spacing to support both the external cabling and the internal RF interconnections.

4.2.10. Conclusion of the hardware design

The hardware design of this project is based on the integration of high-power RF switching and embedded control electronics into a compact two-board system. The design includes a relay-based switching matrix, ground-clamp functionality, directional couplers for power and SWR monitoring, hardware-based frequency detection, and a CYD-based control platform with expanded I/O and analog acquisition.

A major focus of the hardware design was the balance between practical implementation and RF performance. This required careful relay selection, short RF paths, strong grounding, appropriate measurement circuitry, and clear separation between RF and control electronics. The resulting hardware architecture provides a realistic basis for the realization and testing of the final prototype.

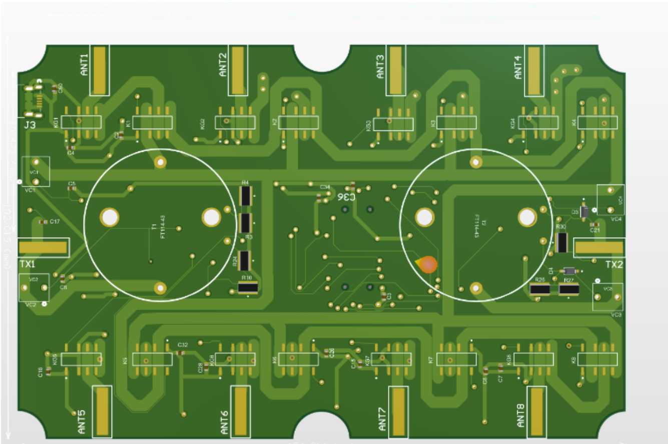

Figure 21. RF board 3D layout

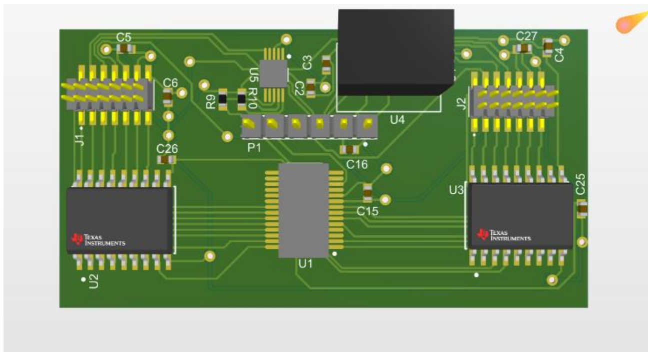

Figure 22. Control board 3D layout

4.3. **Software design**

The software of the antenna matrix is responsible for managing the hardware, processing measurement results, and interacting with users both locally and remotely. Although the hardware provides the switching and measurement platform, the software determines how the system operates, which safety constraints are applied, and how the user interacts with the device. The platform itself consists of the ESP32 microcontroller and a local display. Therefore, the software must control relay matrices, perform analog and digital measurements, manage safety and interlocks, provide local interaction options, and handle remote communications with the Android application.

4.3.1. Software design general considerations

Software may consist of components such as:

- Relay control,
- Safety and interlocks,
- Power and SWR management,
- Frequency measurement,
- Local user interface,
- Remote communication,
- Watchdog supervision.

All those software elements are dependent upon one another; the task of software design is to create a single coherent control system out of the above functionality.

4.3.2. Relay control

One of the key components of software is relay matrix control. As there are many relay matrices in the project, the CYD unit does not directly drive them but rather sends commands to the MCP23017 I²C I/O expander; the MCP23017 provides the control outputs, which are then applied to the ULN2803 driver stages.

For each antenna, the software controls two different relay functions:

- Selector relay, connecting the antenna to one of two transceiver paths;
- Ground clamp relay, grounding the antenna when it is not in use.

Consequently, software must define valid antenna states and translate them into appropriate relay control commands.

4.3.3. State-based switching

To minimize the chances of switching an antenna to an incorrect state (with potentially disastrous effects), the relay control is implemented as state machine; the software guarantees that any antenna is in one of three possible states:

- Connected to the transceiver Bus 1;
- Connected to the transceiver Bus 2;
- Parked (that is grounded).

Illegal states, such as simultaneous connection to both transceiver Buses or grounding of an already-connected antenna are not allowed by the software.

4.3.4. Safety and interlock logic

Safety is a key issue in this project; since the matrix is designed to operate at high power levels and remotely, the software should anticipate potential hazards and avoid rather than react to them.

Software must therefore implement the following safety-related functions:

- Prevent invalid relay states;
- Not allow switching when transmission is in progress (this applies to relay switching, not frequency selection);
- Ground antennas not currently in use;
- Provide SWR warning depending on forward and reflected powers;
- Fallback behavior in case of error condition.

One of the most important safety measures is the "hot switching" prevention; although the relay matrix may well support such switching electrically, changing the RF path mid-transmission increases the risk of contact arcing and damages. To this end, software must be designed in such a way that it blocks any relay command during transmission and while in an operating state that poses unacceptable risks.

4.3.5. Power and SWR processing

The analog power measuring subsystem produces signals proportional to:

- Forward power on transceiver Bus 1;
- Reflected power on transceiver Bus 1;
- Forward power on transceiver Bus 2;
- Reflected power on transceiver Bus 2;

Those analog voltages are sampled via the ADC channels of the ADS1115 converter. The software must periodically read their values and process them.

Based on forward and reflected power data, SWR value is estimated by the software; it is then used for:

- Displaying to the user;
- Transmitting over remote control channel;
- Warning purposes.

Calibration factors are likely to become known only after construction of the actual device is completed, since they depend heavily on the exact implementation of Bruene coupler and power detection circuitry.

4.3.6. Frequency measurement

The software is also able to measure frequency of the RF signal received; after being sampled, attenuated, and protected against overvoltage, the signal is further converted into the digital pulse train by the comparator stage; the latter may be read by the ESP32 through the dedicated input pins.

Measuring frequency is more complicated than measuring power; it requires clean digital timing, accurate edge detection, and a proper GPIO pin layout on the CYD. The software must cope with the above issues, detect the edges of pulses, and convert them into the frequency estimate. The result may then be displayed and further used for user information or automation.

4.3.7. Local user interface

The local user interface is provided by the CYD itself; the latter provides a local display that serves as the principal control panel. The software must present the current system status to the user, allowing him or her to see:

- The currently selected antenna paths;
- Relay state information;
- Forward and reflected power;
- SWR;
- Frequency information;
- Overall system status or warnings.

The local interface must facilitate the operation of the matrix without resorting to remote control.

4.3.8. Remote communication

As was mentioned earlier, the project provides remote control capability using the Blynk app; the user can monitor the state of the system and issue control commands to it.

The remote communication component is capable of:

- Choosing antenna routing paths;
- Reading measured values;
- Viewing warning or status information.

Given that remote control introduces the risk of user commands sent without supervision, software must apply the same safety rules to remote requests as it applies to commands entered locally.

4.3.9. Watchdog and fault tolerance

Given that the system is meant for remote use, software must be fault-tolerant; consequently, software design includes watchdog supervision. A watchdog allows software to recover from unexpected lockup state.

Also, software should bring the system into a known initial state; in other words, once the device is initialized after startup or restart, it should not fall into any undefined relay state.

4.3.10. Conclusion

The software design of the antenna matrix is intended to transform the hardware into a safe and intelligent control system. It does not only switch relays, but also supervises the operating state, processes measurement data, prevents unsafe actions, and provides both local and remote interaction.

A key feature of the software is that it combines control, measurement, and protection into one coordinated system. This is essential for making the antenna matrix reliable and practical for real-world HF operation, especially in remote use conditions.

5. Results

5.1. Implementation and realization

After the analysis, concept selection, schematic design, and PCB development phases, the project enters the implementation and realization stage. It involves the actual creation of a prototype using the obtained designs.

5.1.1. PCB realization

Once schematics and PCB layout are designed, boards should be ordered for fabrication. In this case, the system will consist of two PCBs:

RF board, which includes the relay matrix, Bruene couplers, and RF frequency pickup circuits, and control board, consisting of CYD-based controller, relay drivers, ADC interface, and support circuitry.

The RF board is implemented as 4-layer PCB. Its layers include: Top layer – RF traces, second layer – continuous ground, third layer – +12V distribution, and fourth – extra routing.
Control board will be a 2-layer PCB where the major part of routing occurs on the top layer, while bottom is primarily used for the ground plane.

The selected design facilitates short RF paths and controlled grounding.

5.1.2. Component procurement

With the completion of the PCB design, component procurement should begin. It includes the following items:

- relays for the matrix,
- ferrite toroidal cores and wire for bruene couplers,
- analog and digital support components,
- comparator and adc devices,
- connectors and power components.

Component selection requires numerous revisions, in particular for the relays due to their package type, current, availability and cost. With the final selection made and accepted, appropriate components will be ordered to move to assembly.

5.1.3. Hardware assembly

Upon obtaining the boards, assembly phase should begin. It implies soldering all components and assembling the prototype prior to testing. At this stage, the plan should focus on implementing basic supporting circuitry and relay control path, then measuring and remaining circuitry.

The project integrates RF hardware and digital electronics. Thus, the gradual assembly process allows detecting errors in soldering, wiring, and power supply connections prior to populating the entire system with components.

5.1.4. Preliminary software preparation

Parallel to hardware assembly, software work should proceed. In advance of the prototype's assembly, certain software functions can already be tested separately.

Namely, such functions include:

- communication with MCP23017 I/O expander,
- relay-control logic through ULN2803,
- basic user interface functions on CYD,
- initial code for further measurement processing.

The parallel approach allows saving time while waiting for the completion of the hardware. The further updates in software development, code modifications, and progress implementation will be available in the project GitHub repository:

https://github.com/voltversa/Intelligent-Remote-Controlled-Antenna-Matrix-for-Shortwave-Radio

5.1.5. Current realization status

In the current phase of the project, PCB design is finalized, required hardware is ordered, and necessary components are ordered. Assembly of the boards has already begun, and the preparation of appropriate software has also been started. Consequently, the project has proceeded to the realization phase from purely design-oriented phase.
Updates about assembly, software development, and implementation progress will be provided in the GitHub repository of the project.

5.1.6. Conclusion of the implementation and realization phase

The phase of implementation and realization bridges theory and practice in the project. Having completed the PCB design and ordered all necessary hardware, as well as started the assembly and software preparation, the project has entered the phase of practical realization. Further actions include assembly completion, testing subsystems, software integration, and evaluation of the prototype performance.
Such updates will be continuously added to the GitHub repository of the project.

5.2. Validation and Testing

The last stage in the development of the prototype is its validation and testing. The purpose of this stage is to validate that the developed hardware and software function appropriately and all subsystems operate reliably.

Given the complexity of the system's implementation and its diversity of functionality, validation and testing will occur progressively, i.e., not the whole system at once.

5.2.1. Strategy for validation and testing

As mentioned above, the proposed validation strategy is incremental, meaning that the testing will take place incrementally, i.e., first of all individually for each subsystem.

Among the goals of the process of validation and testing will be:

proper functioning of the control circuitry, proper switching of relay matrix, successful acquisition of analog data, proper operation of the software logic, and safety of the system operation.

Software-related testing and validation updates will be added to the GitHub repository of the project.

5.2.2. Validation of control and communication

The first stage of validation concerns the digital control chain, including communication between CYD and the MCP23017 I/O expander and working of relay drive circuitry.

Testing objectives will include:

- Proper I2C communication initialization with the expander,
- Correct software controlling of expander outputs,
- Proper driving the ULN2803 stages to drive relay coils.

Without proper operation of the digital chain, the relay matrix would not be operating correctly putting the system into an unstable condition. All software and communication-related results will be added to the GitHub repository.

5.2.3. Validation of relay switching

Once the functions of the previous stages are verified, the switching of relay states can be validated.

Software must provide proper relay configuration to avoid short circuits. Thus, the following validation tasks will be conducted:

- switching one bus line connection per each antenna;
- no simultaneous switching an antenna on and grounding it;
- switching the parking antenna into the grounded state.

Updates of relay-switching tests will be added to GitHub together with software updates.

5.2.4. Validation of analog measurement

Once the switching has been tested, validation of the behavior of the analog measurement chain follows.

It is necessary to check that the outputs of the directional couplers are acquired and converted correctly by the ADC.

Thus, it is necessary to verify the following:

- Successful communication with the ADC,
- Accurate acquisition of the forward and reflected signals,
- Correction detection of the channel number,
- Consistency of the measurement results.

Once these points have been confirmed, the next step will be the calibration of the measurement chain in order to obtain meaningful power and SWR values. Because of the practical nature of the Bruene coupler implementation, the final behavior of the measurement subsystem must be verified experimentally.

All further updates related to power reading, SWR estimation, and measurement calibration will be documented in the GitHub repository.

5.2.5. Frequency measurement validation

Next, the frequency measuring subsystem should also be validated as an independent part. It will be necessary to verify that the sampled RF signal is converted into a clean pulse train in the comparator stages and that this signal is counted successfully.

The following points are verified:

- the presence of an easily detectable pulse signal at the output of the comparator,
- the digital inputs of the controller receive the desired waveform,
- the software estimates the frequency of the signal correctly.

Since the CYD platform does not provide many GPIO resources, the testing stage becomes essential in this respect as well.

5.2.6. Safety validation

An important requirement for the operation of the system is its safety and stability in operation. In this respect, investigating the safety properties of the system is crucial.

Thus, testing tasks include:

- Preventing invalid relay configurations,
- Blocking of unsafe switching actions,
- Placing unused antennas in a predefined state,
- Correct indication of faults and warnings,
- Checking the operation in a fail-safe mode after restart.

5.3. Current status of the results

At the current stage, the project has entered the practical realization phase. The PCB design has been completed, the boards and components have been ordered, assembly has started, and the first software preparation is in progress. The next major milestone is the validation of the control chain, followed by relay testing, power-reading validation, frequency measurement verification, and further software integration.

Ongoing progress, software development, power-reading results, and testing updates will continue to be published in the project GitHub repository:

https://github.com/voltversa/Intelligent-Remote-Controlled-Antenna-Matrix-for-Shortwave-Radio

6. Interpretation of results

So far, it should be noted that the project has moved beyond the purely theoretical part to the practical implementation stage. All stages of design (schematic design, PCB design, and component selection) are finished, and currently, the project reaches the implementation phase (PCB manufacturing, component procurement, prototype assembly, and software). It is a crucial achievement since it allows concluding that the theoretical prototype was created into the working one.

Certainly, the most crucial result is related to the optimization of hardware architecture. In the course of development, it became clear that some important design decisions would need to be revised regarding the relays choice, ground clamp logic, and measuring technique. The fact that multiple relay selections had to be performed shows the importance of making such an aspect carefully since relay selection cannot be performed according to the criterion of simplicity. Calculations of voltage and current under mismatch conditions made it clear that significant electrical margins must be provided at the switching stage.

It becomes clear from the design that an optimal design decision for switching in a high-power HF system cannot be considered as using very small signal relays. Another key result is related to the development of hardware architecture as the two-board one with explicit separation of the RF section and the control one. Such a decision as well as the use of the 4-layer RF board and the corresponding grounding of the control board can be regarded as an answer to the interpretation of the project both as a problem of an embedded system and of RF electronics.

As concerns power measuring, the use of a Bruene coupler as the measurement method can also be seen as an engineering compromise decision. Indeed, although more advanced technologies were used, the most realistic option for the thesis prototype was selected (the implementation of such a coupler proved to be relatively easy, tuning could be carried out without any problems, and it is more realistic). The use of hardware-based frequency measuring can also be considered as such a compromise decision to reduce the dependence on the used radio protocol.

Now, at the current stage, the project can boast results proving consistency and feasibility from a technical perspective. However, further development will help to determine how much the final prototype will meet all the set requirements. Future updates related to software development, testing, and power reading validation will be presented in the project GitHub repository.

To summarize, the results achieved so far allow speaking about the successful shift from theory to practice. Also, the results prove that the final quality of the prototype will depend on many factors apart from the initial theoretical concept of the design.

7. Discussion

The realization of the project involved making several engineering decisions regarding the trade-offs between the performance, feasibility, cost, and realization. Therefore, the resulting design is far from being the best theoretical solution in all aspects; rather, it represents a combination of compromises for a prototype of a bachelor thesis project.

One of the main design decisions was the choice of relay devices. The optimal solution for RF switching is using dedicated high-power coaxial RF relays due to their specialized features including low insertion loss, good control interface, and RF power capability. However, those relay types are relatively expensive, and thus the choice has been shifted to more affordable devices. It was followed by several iterations regarding the relay type selection based on its performance under various scenarios until the more realistic choice became obvious.

Another major decision relates to measurement system design. While the tandem coupler can offer higher directivity when comparing to the Bruene device, the latter is preferred due to its simplicity for implementation purposes. The Bruene coupler is therefore believed to provide the functionality necessary for this application at a lower implementation cost and within a more restricted design scope.

Finally, a compromise between using the hardware-based and CAT controlled approach for frequency measurement can be observed in this project design. Using CAT protocol involves additional costs due to dependence on transceiver communication capabilities. On the contrary, designing a hardware solution introduces new requirements in software implementation and increases the number of GPIOs needed for communication with other circuits.

The separation between RF board and control board is another decision of importance. This decision is dictated not only by improvement of shielding and grounding, but also the fact that the project is both an RF switching project and an embedded system development issue. Thus, the usage of 4-layer RF board and ground plane in control circuits is reasonable.

So far, all design decisions seem to be technically justified; however, the actual assessment of their appropriateness can be done after the test and calibration phases. Some factors may affect the quality of prototype performance including

- Switching ability of the relay matrix;
- Accuracy of power measurements;
- Effectiveness of safety logic operation;
- Software implementation.

It is essential to underline that the project itself should not be seen as a final industrial solution, but rather as a development process of the intelligent antenna matrix prototype. Some decisions were made based on the fact that the most optimal choice within limited resources has been used at each step.

Summing up, it is possible to defend the project solution as a feasible and technically sound concept of developing a controlled antenna matrix for the shortwave radio range. Thus, it was demonstrated that it is feasible to design such a system, and the most difficult parts of the design process remain.

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Annexes

The annexes contain supporting technical material that complements the main thesis text. These annexes provide additional detail on the design, implementation, and testing of the system without interrupting the readability of the main report.

Annex A – Full RF board schematic
Complete schematic of the RF board, including the relay matrix, ground-clamp relays, Bruene couplers, and frequency pickup circuits.

Annex B – Full control board schematic
Complete schematic of the control board, including the CYD-related circuitry, MCP23017, ULN2803, ADS1115, comparator connections, and power supply section.

Annex C – PCB layouts
Top, bottom, and additional relevant views of the RF and control PCB layouts.

Annex D – PCB stack-up and grounding
Overview of the RF board 4-layer stack-up and the grounding strategy of both boards.

Annex E – RF and relay calculations
Voltage, current, and mismatch calculations used for relay and PCB design justification.

Annex F – Bruene coupler calculations and component sizing
Supporting calculations and design considerations for the directional coupler implementation.

Annex G – Microstrip calculations
Trace-width calculations and assumptions used for the RF bus design.

Annex H – Frequency measurement circuit details
Additional information related to the RF pickup, attenuation, protection, and comparator stages.

Annex I – Software excerpts
Relevant software fragments used for communication, relay control, measurement, and subsystem testing.

Annex J – Test setup and measurements
Additional screenshots, measurements, and observations obtained during hardware and software validation.

Annex K – Enclosure and mechanical drawings
Mechanical drawings and machining plans for the Hammond enclosure and RF connector placement.

Annex L – Bill of materials
Overview of the main components used in the prototype

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